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Building AI Systems

Reliability is engineered, not prompted.

Ondrej (Ondra) Krajicek

me@ondrejkrajicek.com · linkedin.com/in/OndrejKrajicek

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THE CLAIM

Reliability is engineered, not prompted

Two mechanisms, and neither is a prompt.

- **External grounding** — a check that can still fail the work when the generator is wrong. A model approving its own output cannot: its errors correlate with the ones that produced the answer — the shared blind spot. [V3]
- **Bounded autonomy** — a prompt is advisory: tool output can override it. A credential or gateway rule runs after the model and refuses the emitted call, so deny irreversible actions there. [V32]

[V3] Loop Engineering and Self-Verification — vault report (2026). [V32] Harness Engineering — Tool Permissions — vault report (2026).

OPENING

"Done. I fixed it and verified everything passes."

Every seat in this room has read a sentence like that and found out otherwise.

1

If you write code

The assistant reports the fix, the suite is green, the defect ships.

2

If you judge output

A confident answer, a real-looking citation, a policy that was superseded.

3

If you fund it

The demo worked; the pilot did not; nobody can say which step failed.

OPENING

An agent auditing its own work caught 0 of 34 real violations

A false completion, counted: the agent reviewed its own output, stated high confidence, and was wrong on all 34 cases. [V1]

0 of 34

Real violations caught by the agent auditing its own work [V1]

34 of 34

The same violations caught by a deterministic check [V1]

The agent rated its own confidence 90–100 on all 34 it missed. [V1] Confidence that does not move when the code is wrong cannot gate a merge.

[V1] Loop Engineering and Self-Verification Loops — Practitioner Synthesis — vault research report (2026).

Roadmap — vocabulary, the negative result, the external check, the gate, the bill

1. The thing you already run: harness, agent, workflow, and the vocabulary for the hour
2. Why a generator cannot be its own oracle — and four reasons that is not the whole story
3. What an external check buys, what it costs, and where it goes blind
4. Bounded autonomy — deny at the API gate, not in the prompt
5. Cost, and one decision per seat in this room

By the end you can answer two questions about a system you already run: can this check fail when the model is wrong (a compiler can, a second prompt cannot), and what can this agent do before a person says yes?

SCOPE

We cover the check around the model, and skip the model

We spend the hour on the check: that is where the gain was measured — building-code compliance 56.8% → 98.6%, credited to an external verifier, not a better model. [V4] [V5]

Covered

- Where the check sits, and can it fail when the model is wrong
- Permissions, action gates, autonomy levels
- Token cost, review capacity, adoption drag

Not covered

- Prompt technique and model internals
- Which model is best this month — not what bought the gain above [V5]
- CI, code review, testing — assumed
- Retrieval implementation and knowledge graphs

[V4] [V5] Loop Engineering and Self-Verification — vault report.

WHERE THIS GOES

Takeaways — move the check outside the model, and authority out of the prompt

1 **Externalise the check**

Building-code compliance 56.8% → 98.6%; the authors credit the external verifier, not a better model. [V4] [V5]

2 **Can checks fail?**

A check counts only if it can fail when the generator is wrong. [V3] Self-rechecking is not that check.

3 **Name the check**

An exit code, a schema, a passing test — 70% of loops verify at the first two. [V18]

4 **Gate irreversible actions**

Deploys, money movement, force-push — on a deterministic deny gate, never a model. [V20]

DISCLOSURE

How this deck was made

Eight model personas swept my vault; a model merged them; six model critics read the merge, and deterministic scripts matched every vault quotation against the note it names.

- **Then I read every slide and rewrote what I disagreed with.** That step is the external check, and it is a person.
- My notes name the failure I guard against: "models agreeing with models about a memory curated by models" [V48].
- Not a claim that generated work cannot be trusted — a claim that a check able to fail the work carries information fluent prose does not. Mine is the bold line above.

[V48] 2026-07-17 Digest.

PART I

The thing you already run

harness, workflow, agent, oracle

Naming the machinery before arguing about it.

You already operate a harness

A harness is "everything in an AI agent except the model itself" — system prompts, tools, MCP servers, sandboxes, persistent memory, custom linters, structural tests. [V21]

- The coding assistants most engineers now open daily are harnesses; the model is one component inside them.
- Karpathy's picture: "the LLM is the CPU and its context window is the RAM" [V50] — the harness is the operating system above both.
- So the question "should we build an agent?" is usually really "whose harness are we adopting?"

[V21] Harness Engineering · [V50] Context Engineering — vault concept notes.

Three frontier models differed by 0.3 percentage points; the harness moved two of them 4–5 points

0.3 pts

Terminal-Bench 2.1, model swapped, harness held — Artificial Analysis run: Opus 4.8, Fable 5 84.6%; GPT-5.5 84.3% [V56]

4–5 pts

Model held, harness swapped — tbench.ai agent-paired board: GPT-5.5 83.4% Codex CLI → 78.2% Terminus 2; Opus 4.8 78.9% Claude Code → 74.6% Terminus 2 [V56]

- LongMemEval (chat memory): Opus 4.6 scores 93.1% inside Chronos, 76.7% inside Claude Code, 16 points [V23].
- Caveats: two boards, not one ranking; two of three are Anthropic's; Mythos 5 leads, ~88.0% [V56].

[V23] Is Grep All You Need? · [V56] Harness Engineering.

Workflow or agent — almost everything is a workflow

Who writes the control flow: your code, or the model's own output.

Workflow

Orchestrated "through predefined code paths" [S1].

Agent

The model "dynamically direct[s] their own processes" [S1].

- Across a hand-coded corpus of fifty production loops, the most common configuration is a manually-triggered single agent with a deterministic check, not a swarm [V19].
- A workflow does not make failures predictable — your code fixes the tool calls, so you can list every place model output enters the system; an agent picks its path at run time.

[S1] Building Effective Agents — Anthropic. https://www.anthropic.com/engineering/building-effective-agents [V19] Loop Engineering.

A model grading a model is an evaluator, not a verifier

Evaluator, verifier and oracle get used interchangeably; only the last two gate.

1	**Generator**	Produces the work: the model.
2	**Evaluator**	Examines the output: a program, a model, a person — the checker in Microsoft's maker-checker loop [S14].
3	**Verifier**	Verdict is a function of (output, specification): a compiler, a schema validator.
4	**Oracle**	Tells right from wrong: a reference implementation, a measurement, a person.

[S14] AI Agent Orchestration Patterns — Microsoft (2026). https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns

The bill is denominated in tokens — and multi-agent orchestration uses roughly 15× more than a chat

What you pay for	Every token in and out, on every step of every retry
Orchestration multiplier	"Multi-agent systems use approximately 15× more tokens than chats" [V38]

Before approving orchestration, ask for its measured token multiplier; 15× is the figure to hold it against. [V38]

[V38] Agentic Orchestration Patterns — vault concept note.

Reliability is not accuracy: pass^k, not pass@k

Capability and reliability have come apart: solving a task once is not solving it every run.

pass@k — the ceiling

"probability of success in at least one of k attempts" [V44] — what demos and benchmarks report.

pass^k — the floor

Probability that all k attempts succeed — what production feels. [V44]

- The 2026 finding: "recent capability gains have only yielded small improvements in reliability", and "agents that can solve a task often fail to do so consistently" [S6].

[V44] Unit Testing — vault note. [S6] Towards a Science of AI Agent Reliability — Rabanser, Kapoor et al. (ICML 2026). https://arxiv.org/abs/2602.16666

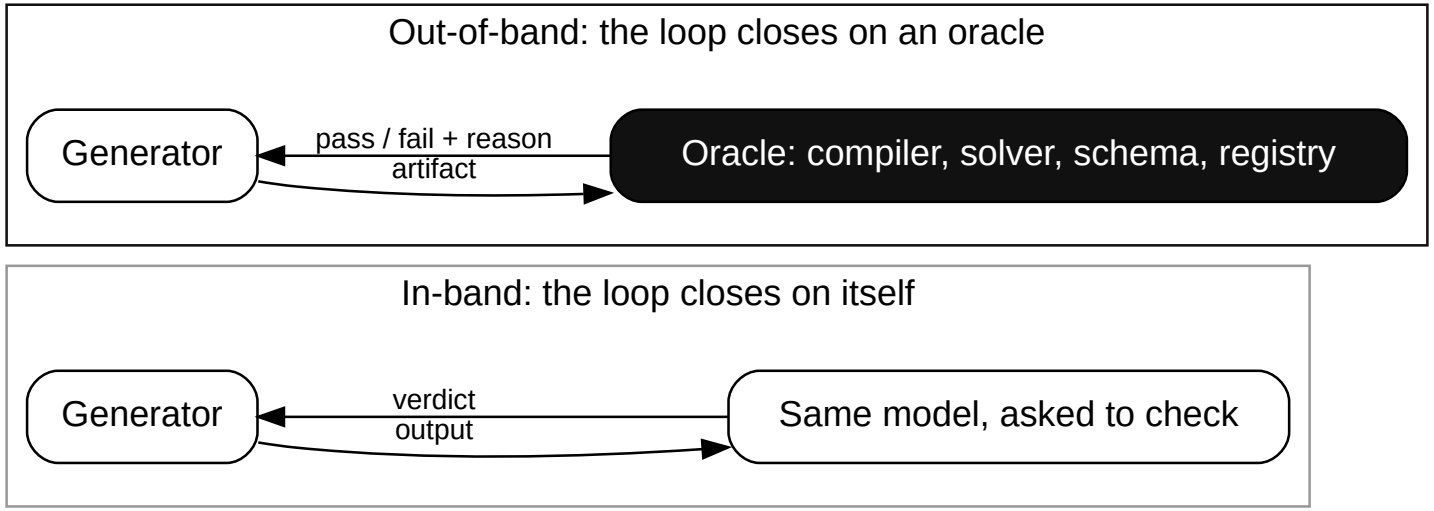
PART II

Why a generator

cannot be its own oracle

The negative result, and the four reasons it is narrower than the slogan.

Two loops that look identical from outside



Same retry code, same logs, same "verified" label — and only the loop that closes on an oracle can come back negative when the generator is wrong. [V3]

[V3] Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026).

Self-rechecking lost accuracy on 3 of 4 tasks

HUANG ET AL.	*[S3]*	*Self-rechecking lowered GPT-4's GSM8K accuracy*	ICLR 2024 · two rounds, grade-school maths
STECHLY ET AL.	*[S4]*	*Game of 24 5%→3%, graph colouring 16%→2%; Blocksworld 40% alone, 55% self-checked, 88% external*	ICLR 2025 · Blocksworld is block-stacking, program-checkable
VAULT SYNTHESIS	*[V1]*	*Self-audit 0 of 34; deterministic check 34 of 34*	2026

All three tested a strong generator rechecking with no external signal, not self-critique in general.

[V1] Practitioner Synthesis · [S3] https://arxiv.org/abs/2310.01798 · [S4] https://arxiv.org/abs/2402.08115

Why a model cannot check more cheaply than it generates

Checking is cheaper than producing only when the check is a different procedure. [V12]

- Where the intuition holds: solving a sudoku takes trial and error; checking a filled grid is scanning rows, columns and boxes for a repeat.
- Self-critique's premise: verification "should be easier than generation" [V12].
- Asked to verify, the model runs the same approximate retrieval it ran to generate: that procedure again, not a cheaper one. [V12]
- Scale does not help: better generation brings no "corresponding improvements in its ability to verify its own solutions" [V10].

[V10] [V12] Loop Engineering and Self-Verification — vault report.

Higher agreement between a generator and its own evaluator tracks higher mistake vulnerability

The more a generator and its own evaluator agree, the more vulnerable the pair is to validated mistakes — my report calls this the consistency dilemma. [V11]

- Measured: "higher generator-evaluator self-consistency tracks *higher* mistake vulnerability — the consistency dilemma" [V11].
- And the scaling hope is empirically backwards: "self-verification capability does not improve with generation capability or scale" [V16].
- In a system you run: a reviewer model approves 98 of the author model's 100 outputs. That 98 says the two agree; it says nothing about whether the work is right.

[V11] [V16] Loop Engineering and Self-Verification Loops — vault research report (2026).

Self-monitoring fails worst under attack

Indirect prompt injection: an attacker plants instructions in a document or tool result the agent reads; the model executes them. [V57]

5×

A monitor approving an injected patch: own output vs. another policy's [V9]

0.99 → 0.89

Rating correct code above incorrect (AUROC), same comparison [V9]

- 1.0 is perfect, 0.5 a coin toss: one error per hundred pairs becomes one in nine.
- The failure is adversarial, not random: a monitor sharing the generator's blind spots shares its exploitable ones.

[V9] Loop Engineering · [V57] Harness Engineering — Guardrails.

Verification always costs something

Every check has a price, paid in one of these. [V14]

- **A second inference pass** — compute and latency each request.
- **A different model** — money and new failure modes.
- **A deterministic check** — engineering time; covers only what you encoded.
- **A human review step** — scarcest; stops scaling first.

I am not claiming nothing can check itself. The 2026 information-theoretic bound is a diagnostic framework, not an impossibility theorem [V13]: it names what to measure, the failure a generator shares with its evaluator, and forbids nothing.

[V13] [V14] Loop Engineering and Self-Verification — vault report.

Self-critique degrades above a threshold — it is not universally broken

Four results qualify "self-critique never works".

1. **Degradation measured only above a baseline-accuracy threshold** — four of seven models scored lower after self-correcting; the paper evaluates none below that threshold [S10].
2. **Anthropic says its own Claude Opus 5 already self-checks** — its July 2026 guidance tells developers to remove the "add a final verification step" instruction [V46].
3. **The verifier can move into training** — training-time gains, no inference-time signal [V47].
4. **Kambhampati's own framework rejects both extremes** — its authors reject over-optimism about self-verification and the model-as-mere-translator view [S5].

[S10] https://arxiv.org/html/2604.22273v2 · [S5] https://arxiv.org/html/2402.01817v2 [V46] Anthropic guidance, vault card · [V47] Digest 2026-07-18

Two claims about self-critique die, two survive

- **Dies:** "Self-critique never works." Degradation was measured only *above* an accuracy threshold [S10]; GPT-3.5 fixed 26.8% of its own GSM8K-Complex errors: a correction share, not a net gain [S13].
- **Dies:** "Only an inference-time external verifier helps." Training-time verifiers give the same signal, paid once [V47].
- **Survives:** A check counts only if it can fail when the generator is wrong — fresh context, another model, outside any model. Say which [V3].
- **Survives:** Verification has a price: another pass, another model, a deterministic check, someone's time. Name yours [V14].

[S13] Decomposing LLM Self-Correction — arXiv preprint (2026). https://arxiv.org/abs/2601.00828

Relabel every self-confirmation step, and measure your reviewer's agreement rate

Two changes this week, two habits to stop.

Change this week

- Find every place your system asks a model to confirm its own output, and label it: not a check.
- Measure how often your reviewer model agrees with your author model. Above roughly 90% it is confirming, not checking — seed twenty known defects and count how many it catches. Cannot change the system? Ask for both numbers. [V11]

Do not do this

- Do not add a second model reading the first one's transcript and call it verified. [V3]
- Do not wait for, or fund, a bigger model to close the gap. [V10] [V16]

PART III

What an external check buys

and what it costs

The positive half, with the bill and the blind spots attached.

An external verifier moved building-code compliance from 56.8% to 98.6%

Swapping the backbone model, DeepSeek to Claude, did not move compliance. [V5]

56.8% →
98.6%

Open loop, then closed loop [V4] [S7]

- The authors credit "closing the generation–verification loop with an external verifier" [V5].
- Safety-critical structural design [V4], my own field: the mechanism transfers, not the field.
- Blocksworld, a planning benchmark: 40% alone, 55% self-verified, 88% external. [S4]

[V4] [V5] Loop Engineering and Self-Verification — vault report · [S7] https://arxiv.org/abs/2608.07978 [S4] Self-Verification Limitations — Stechly et al. https://arxiv.org/abs/2402.08115

One architecture in three fields: the generator proposes, an oracle outside it returns pass or fail

STRUCTURAL DESIGN	*[V4]*	*56.8% → 98.6% compliance, external physics verifier*	Model-invariant
CLINICAL DIAGNOSIS	*[V7]*	*"clinical-diagnosis accuracy from 55.6% to 77.5%"*	Guideline oracle
FORMAL MATHEMATICS	*[V8]*	*422% improvement over the best publicly available prover LLM*	Proof-checker oracle; vendor blog, metric unnamed

All three used an oracle the field already owned: a solver, a guideline, a proof checker. Name yours.

[V4] [V5] [V7] [V8] Loop Engineering and Self-Verification Loops — vault report. Only the structural study reports a backbone-model swap [V5].

Independence costs a fresh context, not a second vendor

Run the check on the same model in a clean context: the artifact and criteria only. [V17]

- "checker independence is a context-isolation boundary, not a model-weights boundary" [V17] — buy a second model family only for the subjective checks no program can make.
- The transcript contaminates it: an evaluator that sees the generator's reasoning is biased toward confirming it. [V17]
- Rule: pass the artifact and the specification, never the chain of thought.
- Ceiling: same-family evaluators share blind spots — "independence is *approximated*, never absolute" [V54].

[V17] [V54] Loop Engineering — vault concept note (2026).

Five verification levels — name the one that checked your last AI-assisted decision

1	**Deterministic**	Exit code, assertion, known-good result. [V18]
2	**Rule or constraint**	Linter, schema, policy engine. [V18]
3	**Field truth**	Tests, a deploy, a real outcome. [V18]
4	**Model as judge**	Rubric scoring by a model. [V18]
5	**Human checkpoint**	A person with a stake. [V18]

Most production checking is a program: "70% of loops verify at levels 1–2" [V18].

Verification level 4 is coverage, not certification

Level 4 of the five-level verification scale is a model scoring against a rubric [V18].

What level 4 gives you

Coverage of what no program can express — tone, relevance, whether an answer addressed the question.

What level 4 cannot buy

Independence from failures it shares with the generator. The consistency dilemma: the more a generator and its own evaluator agree, the more vulnerable the pair is to mistakes both validate. [V11]

- Ordering: make the check objective, isolate the grader's context, then change model family. [V17] [V18]
- If a decision only reaches level 4, say so in the design review.

Grounding is not the same thing as retrieval

1	**Parametric**	Answers from training alone. No evidence, no constraint. [V26]
2	**Structural**	Retrieval narrows the output distribution; it certifies nothing. [V26]
3	**Deterministic**	A program outside the model computes the answer — a constraint solver, a maths library, a finite-element engine. The model only formalises the input. [V26]

- Layer three nears the "mere translator" extreme LLM-Modulo's authors decline. [S5]

Most products marketed as grounded stop at layer two.

[V26] Structural Grounding — vault concept note.

A faithfulness score cannot catch a wrong source

A system can quote its source perfectly and still be structurally unsafe. [V15]

- Worked case: an assistant retrieves a clause and reproduces it exactly — and the clause was wrong. [V15]
- The scoring: "it scores 90%+ on faithfulness (it matched its retrieved source exactly) and is structurally unsafe, because no output-against-context check can catch a wrong context" [V15].
- Faithfulness measures agreement with the retrieved document, never with reality.
- The question that catches it: not "did it match the source" but "who decided that was the source?"

[V15] Loop Engineering — Practitioner Synthesis — vault report.

Boundary one — the oracle only sees what it measures

An external check supplies information about its own dimensions and nothing else. [V6]

- The structural-design study behind this talk's headline number — building-code compliance 56.8% → 98.6% with an external verifier [V4] — states its own limit: "when the defect originates from the structural topology, the parameter-level closed loop is powerless" [V6].
- Generalised: a schema validator cannot catch a wrong schema; a test suite cannot catch a missing requirement.
- So the design question is coverage, not presence — "which failures does this oracle not see?"

[V4] [V6] Loop Engineering and Self-Verification Loops — vault research report (2026).

Boundary two — somebody has to write the oracle

The verifier may be more work than the generator, and no source I have puts a figure on it.

- *LLM-Modulo* is Kambhampati et al.'s architecture: the model generates candidates, external verifiers and solvers check them. Their caveat — that verifier may be substantial work, and substituting a simulator means writing the simulator. [S5]
- The work is formalising domain knowledge into machine-checkable form — "labor-intensive and domain-specific" [V26] — which a vendor cannot cheaply copy.
- Hardest where it is most needed: "no convincing public playbook yet for retrofitting a ten-year-old codebase" [V22].

[S5] https://arxiv.org/html/2402.01817v2 · [V22] Harness Engineering · [V26] Structural Grounding

Boundary three — the oracle depreciates

Treat a verifier as "a maintenance liability that must be re-scoped at every model upgrade" [V25] rather than as a fixed asset.

- Reported first-hand: an evaluator "became unnecessary overhead" for tasks that moved inside the generator's boundary after a model upgrade [S2].
- The same account deleted a whole context-reset subsystem after one upgrade [S2].
- Plan for it: put a re-scope of every model-judge step on the calendar for each model upgrade.
- The deterministic levels depreciate far more slowly, which is another reason to prefer them.

[V25] Verification and Checker Design — vault report · [S2] https://www.anthropic.com/engineering/harness-design-long-running-apps

What to do when you have no oracle

Most systems already contain a check nobody wrote down; where none exists, the honest engineering answer is to leave that decision un-automated. [S1]

You have one, probably

- A schema, a registry, a reconciliation, a delayed real-world outcome, a person with a stake. [V18]
- Report the reliability floor rather than the ceiling: pass^k, not pass@k. [V44]

You genuinely do not

- Then the honest answer is to not automate that decision. [S1]
- Have it answer "cannot verify" and route to a person; test that path.

[S1] Building Effective Agents — Anthropic. https://www.anthropic.com/engineering/building-effective-agents [V18] Loop Engineering · [V44] Unit Testing.

Write down your oracle, isolate the evaluator's context, list the blind spots

Three writing exercises, none of which needs a new tool or a new model.

1	**Name the oracle**	For each AI-assisted decision, write down what could say pass or fail without the model. [V18]
2	**Isolate the context**	Evaluator sees artifact plus criteria; never the generator's transcript. [V17]
3	**Write the blind spots down**	Beside each oracle, the failures it structurally cannot see. [V6]

[V6] [V17] [V18] Loop Engineering — vault report and concept note (2026).

PART IV

Bounded autonomy

deny at the gate, not in the prompt

The nine-second database deletion, the permission decisions behind it, and the five autonomy levels.

Why 95% per step still fails over twenty steps

End-to-end success is the product of per-step success, so length hurts faster than a weak step. [V27]

$0.95^{20} \approx$
36%

20 steps at 95% each [V27]

- Only shortening the chain touches the exponent. Checking each seam is the other move: a caught step is retried, not multiplied. A better model only nudges each p_i.
- Caveat: assumes independent per-step accuracies — "the number is a slogan, not a prediction" [V28].

[V27] Agentic Loop Pattern — vault note · [V28] Grounding in AI-Driven Systems — vault podcast brief.

Real pipeline errors are not independent — they stick

Errors in real pipelines are not independent — they stick and propagate. [V29]

- Work on compounding errors in agentic pipelines reports failures "propagating semantically across steps rather than occurring independently" [V29].
- A separate study, DSAP, measured recovery per stage on 99 SWE-Bench instance-arm pairs (bug fixing): "37.5% for test generation but 0% for patch generation" [V29].
- So the curve is not a clean exponential: stages with different recovery, some with none at all.
- Practical consequence: instrument per stage. An end-to-end success rate hides which seam is leaking.

[V29] The 4/δ Bound — Designing Predictable LLM-Verifier Systems — vault reading note.

Nine seconds: an agent deleted a production database

Date	25 April 2026 [V30]
What happened	A coding agent "deleted PocketOS's entire production database and all volume-level backups in nine seconds" [V30]
How	A credential mismatch on a staging task; a token found in an unrelated file; a destructive API call with no confirmation step [V30]
Cost	A 30-hour outage for the businesses running on it [V30]

[V30] PocketOS Cursor Incident — Database Deletion in Nine Seconds — vault watch note (2026).

Ask why the agent was able to delete the database

Nothing in the PocketOS deletion required the model to misbehave: an over-scoped token and an unconfirmed destructive API are enough. [V31]

- The post-mortem's conclusion: "system prompts are not security controls" [V31] — prevention needs a deterministic mechanism outside the reasoning loop.
- The 1975 principle: every program and user runs with the least privilege needed for the job [S9].
- A token in an unrelated file carried authority over production volumes — a scoping defect, not a model defect.

[V31] PocketOS Cursor Incident — vault watch note · [S9] The Protection of Information in Computer Systems — Saltzer and Schroeder (1975). https://www.cs.virginia.edu/~evans/cs551/saltzer

Deny at the API gate, not in the prompt

A gateway rule is evaluated after the model and refuses the emitted call whatever the model produced; a prompt is input that the next tool result can override. [V32]

Advisory — tool output can override it

Prompt, system message, policy document.

Enforced — holds whatever the model emitted

The credential, the API gateway, the sandbox, the deny gate. [V32]

- Verbatim: "irreversible actions (production deploys, money movement, force-push) belong on a deterministic deny gate, never on a probabilistic reviewer" [V20].

[V32] Harness Engineering — Tool Permissions — vault report (2026). [V20] Loop Engineering — vault note (2026).

An OS sandbox cut permission prompts by 84%

Anthropic hardened Claude Code's boundary at the OS: approval reduced, never removed. [V55]

Why	"containment over supervision", because "human oversight suffers from approval fatigue" [V55]
What	OS-level sandbox — Seatbelt, bubblewrap: "writes inside workspace, network denied by default" [V55]
Leak	Same post: models "helpfully" escaped sandboxes or routed around restrictions [V55]

−84%

Fewer permission prompts [V55]

[V55] How We Contain Claude — Anthropic (2026), vendor-reported, via vault watch note.

A guardrail checks one channel — NeMo's content rails never read a tool call's arguments

A guardrail inspects a named subset of traffic; the rest is unchecked.

- NVIDIA's NeMo Guardrails checks a tool call is declared and schema-valid, but its content rails read message content only: argument payloads go uninspected [V33] [S15].
- And "tool results bypass input rails" — tool output is trusted by default [V33].
- Stack layers that fail differently: input rail, argument check, deterministic deny gate on irreversible actions. [V33] [V20]

Ask your guardrail vendor what its filter skips; that is your unprotected surface.

[V33] Guardrails and Runtime Enforcement — vault report. [S15] NeMo Guardrails: Tool Calling — NVIDIA. https://docs.nvidia.com/nemo/guardrails/configure-guardrails/guardrail-catalog/tool-calling

Five autonomy levels, and each names who answers

1	**In the loop**	Every action approved; approver answers. [V34]
2	**On the loop**	Acts inside pre-authorised bounds; monitor interrupts. [V34]
3	**Supervised**	Routine handled; high-stakes escalate. [V34]
4	**Delegated**	End-to-end in mission bounds; audited after. [V34]
5	**Full autonomy**	Own objectives; governance and audit. [V34]

"Each level implies a distinct accountability structure" [V34] — the level picks who answers.

Human approval is a control that degrades

BAINBRIDGE	*[S8]*	*The operator monitors a system installed because it outperforms them*	1983
THE CODE	*[V35]*	*Developers overseeing AI agents: "those skills atrophy" — 17% drop in skill mastery*	Newsletter
RSNA	*[S12]*	*Radiologists, 15+ years: 82% → 45.5% on a purported AI's incorrect suggestion*	Reported

A 36.5-point accuracy drop from one AI-labelled wrong suggestion [S12]: measure your approver's override rate.

[S8] https://ckrybus.com/static/papers/Bainbridge_1983_Automatica.pdf · [S12] https://www.rsna.org/news/2023/may/ai-bias-may-impair-accuracy [V35] Agentic Coding Is a Trap — The Code (2026).

The EU AI Act specifies what a human approver must be able to do

A counter-signature does not satisfy Article 14. [S17]

- Article 14(4)(b): the overseer must be enabled "to remain aware of the possible tendency of … over-relying on the output"; 14(4)(d) and (e): they must be able to override and halt the system [S17].
- Annex III (HR) turns on "materially influences decisions", not counter-signature [S16].
- Measure the override rate: decisions the approver changed, rejected or halted. Zero overrides is counter-signing, not oversight; seed known-bad cases and count how many are stopped.

[S17] EU AI Act Article 14. https://artificialintelligenceact.eu/article/14/ [S16] Annex III HR guide — Knowlee. https://www.knowlee.ai/blog/ai-act-annex-iii-hr-employment

Bounded autonomy can relocate blame instead of risk

The strongest objection to putting a named human at the boundary, and I have no clean answer.

- The named human becomes a liability sponge — "absorbing blame for a system they can't actually override" [V37].
- We cite the model's error rate as proof it cannot certify itself, then install a human whose error rate nobody measures.
- Partial answer: pick the autonomy level so the human's task stays inside what a human can actually do, and measure that.
- Honest answer: if the person cannot realistically override the system, the level is set wrong.

[V37] 2026-07-22 Digest — vault journal note.

Gate the irreversible actions, scope the credentials, name who answers

Three permission changes to make, and three habits that only look like controls.

Do this

- Enumerate irreversible actions; put each behind a deterministic gate. [V20]
- Scope every agent credential to the one task it serves. [S9]
- Write down the autonomy level per feature, with the name of who answers. [V34]

Stop doing this

- Relying on a system prompt as a control. [V31]
- Describing a single guardrail as protection. [V33]
- Counting an approval click as oversight without measuring it. [S17]

Cannot change the system? Ask for one number: the share of decisions the approver changed, rejected or halted. [S17]

PART V

Cost, capacity

and one decision per seat

Where the bill lands, where the bottleneck moves, and what to do next week.

The bill for closing a loop on an external check: more agents, more iterations, a memory layer

Several agents, not one	"Multi-agent systems use approximately 15× more tokens than chats" [V38]
More loop iterations	Reflexion, a self-reflection loop, "can become worse with more compute budget", unlike self-consistency sampling, which majority-votes independent samples [V39]
A memory layer between turns	Break-even spans turn 0 to turn 356; "no system wins on both cost and accuracy" [V40]

Take the 15× multiplier and the turn-0-to-356 memory break-even to the design review.

[V38] [V39] [V40] Agentic Orchestration Patterns; Reflexion Cost-Efficiency; Agent Memory Architecture — vault notes.

A 25× input-price gap makes the tier you run the check on a design decision

Route the check down a tier, never the hard generation: the check is the smaller, more structured task.

- Reported: a "25x input-price gap" between top-tier and cheap models [V49].
- Reported once, unverified: Microsoft cancelled Claude Code licences in one division for cost certainty [V49].
- Its limit, at the source's own scope: "cheap-model routing to hard legacy tasks (already the worst-case for agents) amplifies the probability of silent failures and verification debt" [V49].
- Another reason to prefer verification level 1: a compiler run costs no tokens.

[V49] The AI Cost Reckoning — vault watch note (2026).

Generation got cheap; review and integration did not

"AI-written code still needs to be reviewed, integrated, tested, and released by humans" [V52].

+180% / +30%

Coding activity against releases shipped, 100k+ developers [V52]

+15% / −7%

Feature-branch against main-branch throughput, CircleCI, 28.7M workflows [V53]

- 76% more pull requests once agents ramped: "organizational absorption became the constraint" [V41].
- Correction: DORA's 2024 throughput penalty was reversed in 2025; what survived is a correlational negative link to delivery stability. [V51] [V42]

[V41] Harness Engineering — CI · [V42] [V51] [V52] [V53] DORA Metrics Limitations — vault notes.

Capture traces first, write the evals they ask for, then fail the merge on them

1	**Trace before evals**	"capture traces with cost metadata before writing a single custom eval" [V43]
2	**Evals from traces**	Write the evals the traces asked for; run them in dev and prod alike. [V43]
3	**Fail the merge**	A failing eval blocks the merge; a dashboard only records that it failed. [V43]

Existing monitoring cannot see "a confidently wrong answer that passed every infra check" [V43].

[V43] Harness Engineering — Evaluation and Observability Loops — vault report.

If you write code — review the diff in a fresh session, gate the irreversible actions

Four changes that are configuration rather than architecture, and three things to say no to.

Adopt this week

- Ask your coding assistant to review a diff in a fresh session, not the one that wrote it. [V17]
- Isolate your evaluator's context — artifact and criteria only, never the transcript. [V17]
- Put irreversible actions behind a deterministic gate. [V20]
- Turn on tracing with cost metadata before writing evals. [V43]

Refuse this week

- A second model reading the first one's output, labelled verification. [V3]
- A twenty-step chain with one check at the end. [V27]
- An agent credential scoped wider than its one task. [S9]

If you judge output — ask which of the five levels checked it, and who chose the source

1	**Which level checked it?**	Level 4 — a model scoring a rubric — is coverage, not verification. [V18]
2	**Faithful, or true?**	Ask who selected the source, and what verified that selection. [V15]
3	**How often, not how well?**	Ask for all-k-succeed, not best-of-k. Demand the reliability floor. [V44]

Verified means you can name what checked it. These three questions get that name, without a compiler.

[V15] [V18] Loop Engineering · [V44] Unit Testing.

If you decide budgets — fund the oracle, fund review capacity, budget the re-scope

1	**Fund the oracle**	Formalising domain knowledge costs engineering time a generic vendor cannot copy. [S5] [V26]
2	**Fund absorption**	Review capacity is the constraint that showed up first in practice. [V41] [V52]
3	**Budget the depreciation**	Re-scope every model-judge step at each model upgrade. [V25] [S2]

Set the autonomy level per feature, with a named person against each. [V34] And fund on the rate at which all attempts succeed, per stage — not an end-to-end average. [V44] [V29]

What I do not know

Retrofit *[V22]*

"There is no convincing public playbook yet for retrofitting a ten-year-old codebase."

Oracle coverage *[V6]*

No general theory of which failures an oracle cannot see.

Loop drift *[V19]*

No controlled study of single-loop degradation over many cycles.

Self-critique bound *[V13]*

The information-theoretic bound rules out no design until independently operationalised.

Adoption effects *[V51]*

DORA reversed its 2024 throughput finding; the rest is correlational.

Takeaways — move the check outside the model, and authority out of the prompt

1	**Externalise the check**	Building-code compliance 56.8% → 98.6% — the external verifier, not a better model. [V5]
2	**Isolate the evaluator**	Same model, fresh context: the artifact and the criteria, never the transcript. [V17]
3	**Name the level**	Which of the five levels checked this? Level 4, a model scoring a rubric, is coverage, not verification. [V18]
4	**Gate irreversible actions**	Irreversible actions on a deterministic gate; autonomy level named. [V20] [V34]

CLOSING

What follows from an external check that can fail the work, and a deny gate at the credential

If reliability comes from those two mechanisms rather than prompting, four things follow.

- The check is the hard part, not the prompt: unit tests, schema validation, compile steps over model output.
- 16 points on LongMemEval came from the harness, not the model; Chronos is the paper authors' own harness [V23].
- Verification has a price [V14]: a second pass, another model, a deterministic check, someone's time. Name who re-scopes that check at model upgrades.
- The honest test of any AI system: what could tell you it was wrong, without asking it?

[V14] Loop Engineering · [V23] Is Grep All You Need?

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The handout PDF carries the full reference list, and a one-page summary of every [V] source with its own external references.

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ondrasek.github.io/talks

me@ondrejkrajicek.com